

Ferroptosis in cancer therapy

A fixture document for extraction drift.

Lipid peroxidation drives an iron dependent death.

GPX4 is the canonical defence against that peroxidation.

System Xc minus imports cystine to sustain glutathione.

Erastin inhibits that transporter and starves the cell.

FSP1 and DHODH repair peroxides in parallel to GPX4.

ACSL4 sets how much oxidisable substrate the membrane holds.

Labile iron drives the Fenton chemistry that propagates.

Ferritinophagy releases stored iron into that labile pool.

PROM2 exports it again, draining the pool and resisting.

The balance of those two decides whether a cell dies.

None of these sentences is a claim about the literature.

They exist to give the extractor enough text to work on.



First block by position, drawn second.

Second block, drawn first in the content stream.

Padding so the extracted document clears the one thousand character floor that the production extractors apply before they accept a PDF at all, since a shorter fixture would sit inside the region that real code discards and could not be passed through the real function under test.

Trailing lines keep the page count and ordering intact.

The reading order property above is unaffected by them.

Each line is plain ASCII in the base fourteen Helvetica.

No ligature, no embedded font, no column layout is present.

Those shapes are named in the test as explicitly uncovered.

A real paper carries all of them and this one does not.

That limit is the honest boundary of what this guard proves.